Weekly Planner

Week #: 11 (09/11/2026 To 13/11/2026)

Subject: Pre-Calculus Math **Unit Title: Functions, Graphs, Transformations, and Inverses** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1 Identifying One-to-One Functions from Graphs	• I can identify one-to-one functions from graphs. • I can use the Horizontal Line Test to explain my decision.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.1 Make goals clear, visible, and supported by short-term objectives.	• Lesson 1.7 PowerPoint and worksheet on Microsoft Teams • Desmos Graphing Calculator
Lesson 2 Finding the Inverse of a Function	• I can find the inverse of a simple linear function algebraically. • I can exchange x and y and solve for y.		
Lesson 3 Graphing Inverse Functions	• I can graph an inverse by exchanging ordered-pair coordinates. • I can explain reflection across y = x.		
Lesson 4 Project Assignment	• I can apply one-to-one functions and inverses in a mathematical project. • I can communicate a plan using correct notation and graphs.		
Homework	Homework: Complete worksheet included on Teams (Lesson 1.7).		
Teacher's notes/ Reminders	• Check Microsoft Teams for Lesson 1.7 resources, worksheet, and project instructions. • Bring a charged device and graphing materials. • Review the Horizontal Line Test, inverse notation, and reflection in y = x.		

Technology Integration	Desmos.com, Microsoft Teams, graphing calculator, GeoGebra, coordinate grids.
Cross-curricular links	**Computer Science:** Reversible processes and input-output mappings. **Cybersecurity:** Encoding and decoding as inverse operations. **Physics:** Reversing simple input-output relationships.
External Assessment Integration	SAT/ACT/AP Precalculus-style graph interpretation, horizontal line test, inverse notation, simple algebraic inverses, and reflection about y = x.
Wellbeing/ ATL Skills	**Critical Thinking:** Justifying whether a graph is one-to-one. **Communication:** Explaining inverse steps and graphical symmetry. **Self-Management:** Planning and completing the assigned project.
Standards	CCSS.HSF-BF.B.4a Find inverse functions. CCSS.HSF-IF.C.7 Graph functions and identify key features. CCSS.HSF-IF.B.4 Interpret functions represented graphically and algebraically.
Inquiry/ TOK Question	How does reflecting a function across y = x help us understand its inverse?
Success Criteria (Measurable)	**Lesson 1: One-to-One Functions** • All students can apply the Horizontal Line Test to simple graphs. • Most students should identify one-to-one and non-one-to-one graphs correctly. • Some students could justify their reasoning using horizontal intersections. **Lesson 2: Finding Inverses** • All students can exchange x and y in a simple linear equation. • Most students should solve for y and write inverse notation correctly. • Some students could check inverses using composition. **Lesson 3: Graphing Inverses** • All students can swap coordinates of ordered pairs. • Most students should plot a function and its inverse accurately. • Some students could explain domain-range interchange and symmetry about y = x. **Lesson 4: Project Assignment** • All students can identify project requirements and select a suitable function. • Most students should prepare a plan showing the function, inverse, and graph. • Some students could explain a real-life application of their inverse pair.

National Identity Domain	Conservation الحفاظ علي البيئة	Global Understanding التفاهم العالمي	Heritage التراث
	Area: UAE Innovation and Future Readiness Area: UAE culture and heritage Area: UAE Economic Development **Area:** Responsible Citizenship and Innovation		
	Implementation: • Students explore inverse relationships in UAE smart technologies and digital services. • Students discuss encryption and decryption in digital security. • Students consider reversible input-output models in UAE innovation and business. • Project planning encourages independent research, communication, and future-readiness.		